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Effect of *Argemone mexicana* (L.) against lithium-pilocarpine induced *status epilepticus* and oxidative stress in wistar rats

G Asuntha[1], Y Prasanna Raju[2]*, CR Sundaresan[3], Arun Rasheed[4], V Harini Chowdary[2], KR Vandana[2], K Satish Babu[2] & KVSRG Prasad[5]

[1]Department of Pharmacy, S P W Polytechnic; [2]Pharmaceutics Division, Sree Vidyanikethan College of Pharmacy, A. Rangampet, Tirupati 517 102, India
[3]Faculty of Medicine, UCSI University, Kuala Lumpur, 56000 Malaysia.
[4]Department of Medicinal Chemistry, Al-Shifa College of Pharmacy, Poonthavanam, Kizhattur 679 322, India
[5]Institute of Pharmaceutical Technology, Sri Padmavathi Viswavidyalayam, Tirupati 517 502, India

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Argemone mexicana (L.) has a role in the treatment of epileptic disorders in Indian traditional system of medicine. We studied its effect on induced *status epilepticus* (SE) and oxidative stress in rats. SE was induced in male albino rats by administration of pilocarpine (30 mg/kg, ip) 24 h after injection of lithium chloride (3 mEq/kg, ip). Different doses of the ethanol extract of *A. mexicana* were administered orally 1 h before the injection of pilocarpine. The severity of SE was observed and recorded every 15 min for 90 min and thereafter at every 30 min for another 90 min, using the Racine scoring system. *In vivo* lipid peroxidation of rat brain tissue was measured utilizing thiobarbiturate-reactive substances. Both *in vitro* free radical nitric oxide and 2,2-diphenyl-1-picryl hydrazyl scavenging activities of the extract were also determined. The SE severity was significantly reduced following oral administration of the extract at 250, 500 and 1000 mg/kg doses. None of the animals from groups 3 to 5 (with *A. mexicana* extract) have exhibited forelimb clonus of stage 4 seizure. The extract also exhibited both *in vivo* and *in vitro* antioxidant activities.

Keywords: Antiepileptic drugs (AED), Antioxidant, Epilepsy, Flavonols, Free radical scavenging, Herbal drugs, Lipid peroxidation, MDA, Mexican prickly poppy, Phytomedicine, Pilocarpine, Seizures.

Persistent seizures lasting over 30 min or two or more seizures without full recovery of consciousness between the seizures is termed as *status epilepticus*. Severe morbidity and mortality are the major neurological consequences of convulsive *status epilepticus*. Most seizures last anywhere from a few seconds to less than 5 min, but in *status epilepticus*, a seizure will persist for 15 min or more[1]. It requires serious medical attention. About 75% of the patients are on monotherapy of antiepileptic drugs for the control of epilepsy. About 5-10% of patients require the addition of other antiepileptic drugs. Another 20% of patients do not respond to therapy[2]. The major causes of convulsive *status epilepticus* are the abrupt withdrawal of antiepileptic drugs (AED), neurologic infections, cerebrovascular accidents, brain tumors, drug toxicity and metabolic disorders. The first line of drugs in the management of *status epilepticus* includes benzodiazepines, phenobarbital and phenytoin. The second line of drugs includes intravenous anaesthetics such as barbiturates, propofol and ketamine[3]. Though these synthetic moieties are effective against *status epilepticus*, they are expensive and many are not affordable. Phyto-medicines have a major role in the development of novel antiepileptic drugs. Many herbal drugs and plants such as *Tephrosia purpurea* (L.) Pers.[1] and *Indigofera tinctoria* Linn[4]. of Fabaceae and *Centella asiatica* of Mackinlayaceae[5] are known sources of compounds which have anticonvulsant activity with few or no side effects[6].

Oxidative stress has a significant role in the pathogenesis of many neurodegenerative ailments. Endogenous antioxidants and enzymes defend against the harmful effects of free radicals. It is a measure of the extent of risk of development of the disease, and to monitor efficacy of therapy. Excessive free radical generation and increase in lipid peroxidation occurs during *status epilepticus* induced by pilocarpine[7].

*Correspondence:
Fax: +91-877 2248226
E-mail: kanishka9002@gmail.com

Argemone mexicana (L.) (Papaveraceae) possess anti-inflammatory, expectorant, emetic, anthelmintic, antipyretic and sedative properties[8]. It is also reported to have antiepileptic activity in some folklore medicine[9]. However, there are no systematic pharmacological studies to support the anti-*status epilepticus* activity of *A. mexicana*. Hence, the present study has been aimed to explore the anti-*status epilepticus* efficiency of ethanol extract of *A. mexicana* plant on lithium-pilocarpine induced *status epilepticus* and its effect on the oxidative stress in Wistar rats.

Materials and Methods

Plant material—The whole plant of *Argemone mexicana* (L.) was collected locally near Srinivasa Mangapuram area, Tirupati, Andhra Pradesh, India during April-June as per to its availability. The plant was authenticated and a voucher specimen was deposited at the herbarium of the Institute of Pharmaceutical Technology, A Women's University, Tirupati, India (Ref: 01/AM/06-IPT-SPMVV/TPT/2004).

Drugs and chemicals—Pentylenetetrazole (PTZ), pilocarpine nitrate and artificial cerebrospinal fluid (ACSF) were purchased from Sigma-Aldrich, USA. Lithium chloride, HPLC grade chemicals and solvents of analytical grade were purchased from SD Fine Chem Ltd., Mumbai, India. Diazepam was obtained from Ranbaxy, New Delhi, India.

Plant extract—The collected plant material was shade-dried at room temperature in the laboratory, powdered and passed through a sieve (coarse 10/44). Plant powder (200 g) was extracted with 1000 mL of 95% ethanol under reflux by heating over a water bath and continued in three batches with the same marc. The extracts were then vacuum dried. The yield of ethanol extract of *A. mexicana* (EEAM) was 30.56% (w/w). The suspension of the plant extract was prepared with 2% Tween 80 before administration to animals.

Experimental animals—Male albino Wistar rats weighing 150–200 g and aged 6-8 weeks of age were housed in polypropylene cages and maintained under standard laboratory conditions with a 12:12 h L:D cycle along with free access to standard rat pellet diet (Lipton, India Ltd.) and drinking water. They were acclimatized to laboratory conditions for ten days before commencing the experiment. The experimental protocol was approved by the Institutional Animal Ethical Committee (Ref. no: 1220/a/08/CPCSEA).

Acute toxicity and gross behavioural changes—Acute toxicity studies for EEAM were carried out according to OECD guidelines. The rats were fasted overnight with free access to drinking water and divided into following six groups of six rats each. Group I animals served as a control and received distilled water orally (2 mL/kg). Animals in Groups II to VI received 0.25, 0.5, 1.0, 2.0 and 4.0 g/kg, respectively, of the EEAM extract by gastric intubation with a soft rubber catheter. The animals were observed continuously for 2 h and then at one-hourly interval for 24 h. Behavioural, neurological and autonomic profiles were recorded at these points of time. The animals were observed for mortality up to 48th hour[10].

Lithium-pilocarpine induced status epilepticus—Thirty rats were divided randomly into following five groups of 6 rats each. Group I animals (control) were administered distilled water (2 mL/kg, oral). Group II animals (reference standard) were treated with diazepam (2.0 mg/kg, ip) while Groups III–V animals were administered 250, 500 and 1000 mg/kg orally of the EEAM extract, respectively. All the rats received lithium chloride (3 mEq/kg, ip) twenty 24 h prior to treatment. One hour after treatment, *status epilepticus* was induced in all the animals (including control group) by administering pilocarpine (30 mg/kg, ip). The severity of *status epilepticus* was observed and recorded at every 15 min until the 90th min and thereafter at every 30 min until the180th min, using the Racine scoring system: 0 (no response); 1 (fictive scratching); 2 (tremors); 3 (head nodding); 4 (forelimb clonus) and 5 (rearing and fall back)[11].

Effect of ethanol extract of A. mexicana (EEAM) on brain lipid peroxidation—The experimental Wistar rats were randomly divided into following five groups of six rats each. Group I were kept as control and were administered distilled water (2.0 mL/kg, oral). Pentylenetetrazole (PTZ) (70 mg/kg) was sub-cutaneously given to the animals of Group II and considered as reference group. Test groups III–V were administered with 250, 500 and 1000 mg/kg extract orally. One hour after the administration of EEAM, PTZ

(70 mg/kg) was injected subcutaneously to all test group animals. On observing the onset of convulsions following the administration of PTZ, the animals (including control group) were sacrificed by decapitation, brains removed and stored. The brains were retrieved from the ice-chamber and submerged in 5 mL of methanol and homogenized and centrifuged at 10,000 rpm at-10°C 15 min for lipid peroxidation assay. The supernatant was assayed based on the reaction between malondialdehyde (MDA) and thiobarbituric acid (TBA)[12]. The *in vitro* antioxidant activity of the EEAM was assessed with scavenging of free radicals such as nitric oxide (NO) and 2,2-diphenyl-1-picryl hydrazyl (DPPH)[13].

Statistical analysis—The data were expressed as mean ±SE *Statistica* (StatSoft Inc.) version 6.0 was used to analyze the data. ANOVA was used for multiple comparisons followed by Tukey–Krammer test. In addition lithium-pilocarpine induced seizures were analyzed using Kruskal-Wallis ANOVA followed by Dunnet's test. P <0.05 was considered statistically significant. The statistical significance was set accordingly.

Results

Acute toxicity and gross behavioural changes—The extract of *A. mexicana* did not provoke any gross behavioural changes or manifestations of toxic symptoms in the animals such as weight loss, increased motor activity, muscle relaxation, hypnosis, tremors, clonic convulsions, spasticity, loss of right reflex, decreased motor activity, tonic extensions, muscle spasm, ataxia, sedation, arching and rolling, lacrimation, salivation, viscid, watery diarrheoa, writhing and urination over a period of 24 h. Ethanol extract of *A. mexicana* was found to be non-lethal even at the maximum single oral dose of 4.0 g/kg. Hence, the doses of the ethanol extracts selected for the study were 250, 500 and 1000 mg/kg orally.

Effect of ethanol extract of A. mexicana on lithium-pilocarpine induced status epilepticus—The results of the anti-*status epilepticus* activity of the extract are shown as scorings in Fig.1. All the animals in the control group exhibited stage 4 seizures within 45 min of pilocarpine injection. The animals in Group II, pre-medicated with the reference standard drug (diazepam) did not demonstrate stage 2 seizures or beyond. The EEAM was able to diminish the intensity of seizures in all test groups of animals and none of the animals exhibited stage 4 seizures. In the EEAM treated animals, a significant (P <0.01, P <0.001) reduction in the severity of lithium-pilocarpine induced *status epilepticus* was observed at oral doses of 250, 500 and 1000 mg/kg. The animals were almost normal in behaviour after 180 min.

Effect of ethanol extract of A. mexicana on brain lipid peroxidation—Pentylenetetrazole (PTZ) induced lipid peroxidation in the brain. The brain lipid peroxidation was displayed with the elevation of brain MDA level in group II animals. The EEAM had significantly (P <0.001) decreased the brain MDA level with dose-dependent manner as shown in Table 1.

In-vitro antioxidant activity—The EEAM exhibited a significant (P <0.01) *in vitro* antioxidant effect against NO and DPPH scavenging in

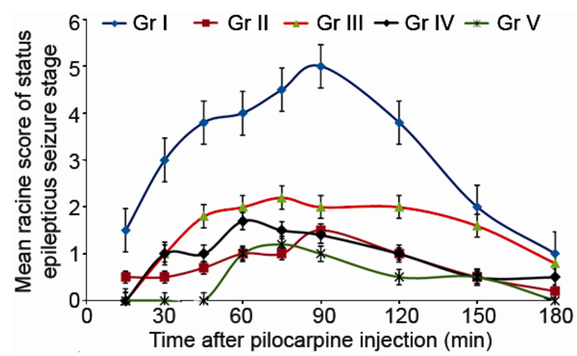

Fig. 1—Effect of ethanol extract of *A. mexicana* (EEAM) on lithium-pilocarpine- induced *status epilepticus*. [Group I control received distilled water orally (2 mL/kg); Group II (reference standard) received diazepam (2 mg/kg, ip); and Groups III–V received 250, 500 and 1000 mg/kg of EEAM extract, respectively. Values are expressed as mean Racine score of seizure stage are mean ± SE (n=6)].

Table 1—Effect of ethanol extract of *A. mexicana* (EEAM) on brain MDA levels [Values are mean±SE from 6 animals each]

Treatment Group (Dose (mg/kg))	Brain MDA level (µM/g)
Control (2.0 ml)	0.87±0.62
PTZ (70)	1.82±0.24[a]
EEAM (250)+PTZ	0.81±0.50*
EEAM (500)+PTZ	0.75±0.48 *
EEAM (1000)+PTZ	0.69±0.53*

*P <0.001 as compared to PTZ treated group. [a]PTZ treated group MDA content was considered as 100% peroxidation

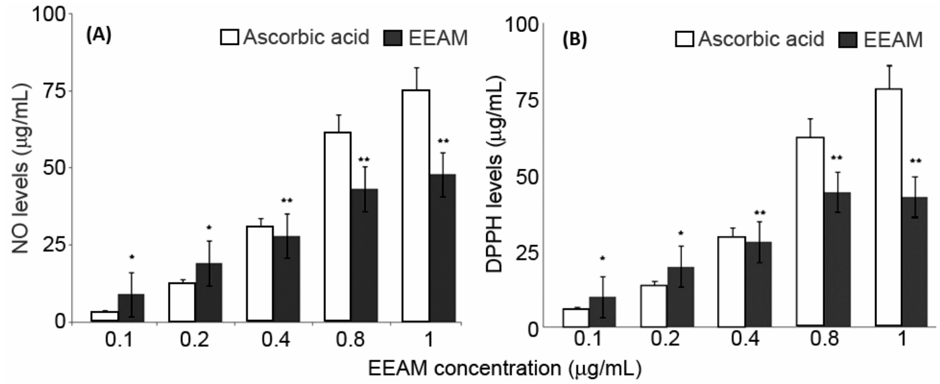

Fig. 2—Effect of ethanol extract of *A. mexicana* (EEAM) against *in-vitro* free radical (A) NO scavenging activity; and (B) DPPH scavenging activity. [Values are mean ± SE (n=6). *P* values: *<0.01; **<0.001 as compared to ascorbic acid].

concentration dependent manner as depicted in Figs. 2 A and B.

Discussion

Seizures induced by pilocarpine, is one of the most frequently used models of temporal lobe epilepsy[14,15]. In this model, lithium does not have any proconvulsant effect in rats[16]. However, the animals that were pretreated with lithium have limbic seizures, following subconvulsant doses of pilocarpine. The combined type of treatment with lithium–pilocarpine results in the accumulation of acetylcholine, inositol monophosphate and the reduction in cortical inositol, and thereby produce ten times greater effects than obtained by either drugs independently[17].

Results of ethanol extract of *A. mexicana* have exhibited a dose-dependent reduction in severity of the lithium-pilocarpine induced *status epilepticus*. Lithium-pilocarpine induced *status epilepticus* correlates with both electrophysiological and neuropathological characteristics of temporal lobe epilepsy[18,19]. Hence, *A. mexicana* can also be useful when administered with other drugs in temporal lobe epilepsy.

Oxidative stress in the brain has been demonstrated in several rodent models of experimental epilepsy such as with PTZ induced seizures. PTZ may also trigger a variety of biochemical process including the activation of membrane phospholipases, proteases and nucleases[20]. Alterations in membrane phospholipid metabolism lead to the liberation of free fatty acids, diacylglycerols, eicosanoids, lipid peroxides and free radicals. Hence, these studies were carried out on ethanol extract of *A. mexicana* for the free radical scavenging activity against the NO and DPPH and their effects on brain lipid peroxidation. The ethanol extract of *A. mexicana* exhibited significant free radical scavenging activity against NO and DPPH and also reduced the MDA levels in the brain.

Our results show that pilocarpine induced *status epilepticus* can produce alterations in different areas followed by neuronal damage. There is an evidence of reduced oxygen availability after *status epilepticus*, both clinically and experimentally. The evidence for the role of free radicals in *status epilepticus* is mostly based on the use of enzymatic and non-enzymatic antioxidant treatment for protection against seizures and for *status epilepticus*-induced neuronal damage[21].

Conclusion

Overall, this study demonstrated that ethanol extract of *A. mexicana* has ability in reducing the severity of *status epilepticus* in Wistar rats. These results support the acclaimed use of this plant in Indian system of medicine. Its anti-*status epilepticus* activity may be due to the presence of polyphenols and flavonoids of *A. mexicana.* Ethanol extract of *A. mexicana* also possessed both *in vitro* and *in vivo* antioxidant activity.

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